



Khwaja Yunus Ali University

Lab Report -03

Name of the Department: Computer Science and Engineering

Course Code: CSE 0713-1103

Course Title: Electrical Circuit Lab

Experiment No.: 03

Name of the Experiment: Varification of Kirchhff's Voltage Law (KVL) and voltage divisor rule using hardware and digital simulation.

Date of Experiment : 26-01-2025

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Instructor Signature & Date

Submitted by –

Submitted to –

Name: Md Sojib Ahmed
ID Number: 06224205101005
Batch No: 18th
Semester: 1st Year 2nd Semester
Khwaja Yunus Ali University

Name: Sakil Ahammed
Lecturer
Department of EEE
Khwaja Yunus Ali University

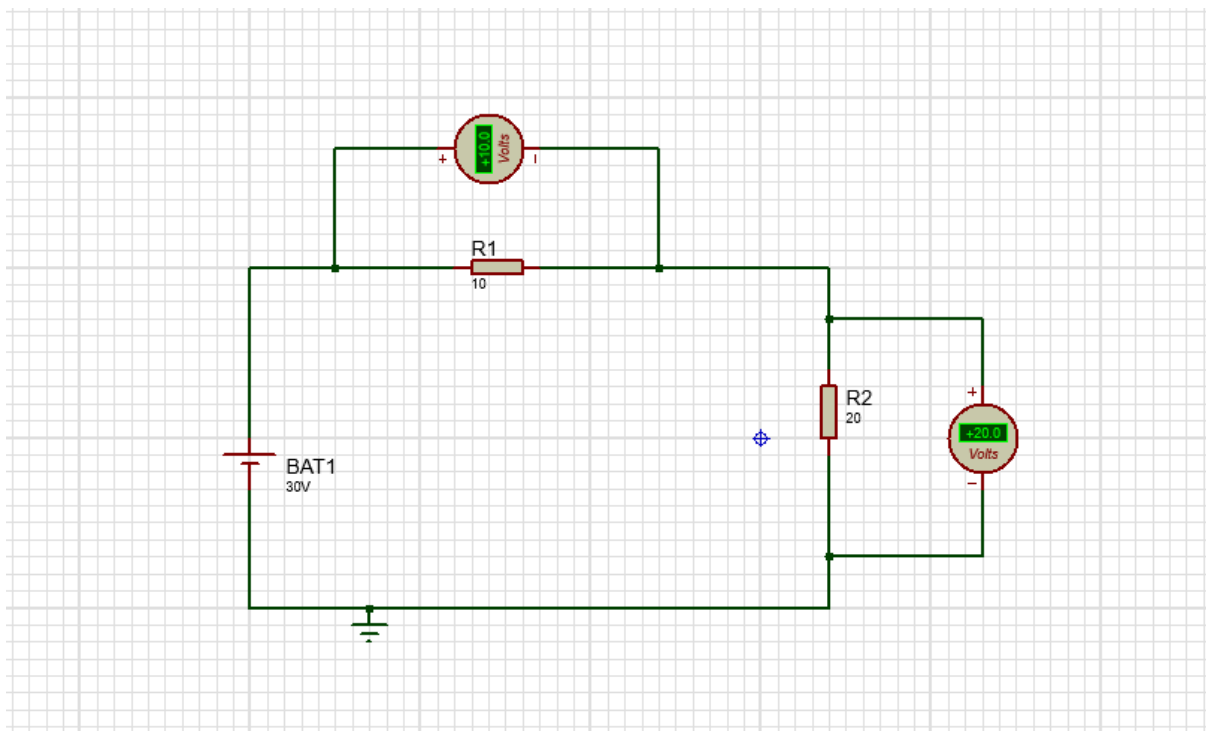
No of Experiment : 03

Name of the Experiment: Varification of Kirchhff's Voltage Law (KVL) and voltage divisor rule using hardware and digital simulation.

Objectives: To verify Kirchhff's villages Law (Kvt) and Vottage dividep pule using hardware and digital simulation.

Theory: Kirchhoff's voltage law: In a cosed electric circuit the algebraic sum of potential drops is equal to the algebraic sum of total electromotive force occuring round the Circuit

Circuit Diagram:



Apparatus:

Software used: Proteus 8 Professional

(i)Voltmeter, (ii)Resiglor,(iii) Power Supply,(iv) Connecting wires

Working procedure:

1. I made the connection as shown in the circuit diagram Using Proteus 8 Professional.
2. I measured the voltage across each resistor.
3. I verified Kirchhoff's Voltage Law (KVL).
 - I measured the voltage across each resistor in the closed loop.
 - I calculated the algebraic sum of the voltages.
 - I verified that the sum was Indeed zero.
 - I measured the voltage across each resistor.
 - I calculated the voltage divider ratio for each resistor.
 - I compared the calculated and measured voltage.
4. I assumed the current flowing through each resistor as 1,12. un
5. I assumed the direction of current flowing through each branch.
6. I measured the current flowing into the top node of the circuit from each of the two branches.
7. I recorded these measured currents.
8. I then the two currents to verify they were zero, or close to zero, confirming that the sum of incoming currents is equal to the sum of outgoing currents.
9. For KVL. I measured V_{R1} , V_{R2} and applied KVL as per the observation table and recorded these measured Voltages.

Calculation: For KVL,

$$\begin{aligned} V_1 &= \frac{R_1}{R_1 + R_2} \times V \\ &= \frac{10}{10 + 20} \times 30 = 10 \text{ v} \end{aligned}$$

$$R_1 = 10$$

$$R_2 = 20$$

$$V = 30 \text{ v}$$

$$V_2 = \frac{R_2}{R_1 + R_2} \times V$$

$$= \frac{20}{10+20} \times 30 = 20 \text{ v}$$

∴ Total valage , V =10+20=30v

Similarly,

$$V_1 = \frac{20}{20+30} \times 120$$

$$=48 \text{ V}$$

$$V_2 = \frac{30}{20+30} \times 120$$

$$= 72 \text{ v}$$

∴ Total valage , V =48+72 = 120v

For KVL,

SL No	Input Voltage (V)	VR1 (V)	VR2 (V)	Total Voltage (V) (calculated)	Total Voltage (V) (Theoretical)
1	30	10	20	30	30
2	120	48	72	120	120
3	180	72	108	180	180
4	240	96	144	240	240

Result and Discussion:

1. The sum of voltage (calculated) is Gov and Voltage (theoritical) is Gov

We have compared the measured values are a little diviated from the theoritical values for calculation and data entry.

Thus the KVL is verified successfully.

Conclusion: In a dosed circuit, the algebric sum of all the EMF and all the voltage drops is zero, hence Kirchhoff's law voltage. is verified.